



Pathway to finding the Kalman gain

- So far, we have derived state estimate $\hat{x}_k^+ = \hat{x}_k^- + \mathbb{E}[\tilde{x}_k^- | z_k]$.
 - But, what is $\mathbb{E}[\tilde{x}_k^- | z_k]$?
- To evaluate this expression, we consider first very generically the problem of finding $f(x | z)$ if x and z are jointly distributed Gaussian vectors.
 - In the derivation, the joint distribution is described using augmented vector X :

$$X = \begin{bmatrix} x \\ z \end{bmatrix}, \quad \mathbb{E}[X] = \begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix}, \quad \Sigma_{\tilde{X}} = \begin{bmatrix} \Sigma_{\tilde{x}} & \Sigma_{\tilde{x}\tilde{z}} \\ \Sigma_{\tilde{z}\tilde{x}} & \Sigma_{\tilde{z}} \end{bmatrix}.$$

- Again, generic x and z are not necessarily the state and measurement in this derivation; think about them (for now) as general RVs.
- After we find the generic result for $f(x | z)$, we will then apply it to our specific problem of determining $f(\tilde{x}_k^- | z_k)$ by setting $x = \tilde{x}_k^-$ and $z = z_k$.
 - We can then use this conditional pdf to find $\mathbb{E}[\tilde{x}_k^- | z_k]$.



Expanding the conditional pdf

- To proceed, we first write $f(x | z) = f(x, z)/f(z)$.
- Note that this is proportional to:

$$\frac{f(x, z)}{f(z)} \propto \frac{\exp\left(-\frac{1}{2} \left(\begin{bmatrix} x \\ z \end{bmatrix} - \begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix} \right)^T \Sigma_{\tilde{X}}^{-1} \left(\begin{bmatrix} x \\ z \end{bmatrix} - \begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix} \right)\right)}{\exp\left(-\frac{1}{2} (z - \bar{z})^T \Sigma_{\tilde{z}}^{-1} (z - \bar{z})\right)},$$

where the constant of proportionality is not important to subsequent operations.

- Recall that $\exp(A)/\exp(B) = \exp(A - B)$... which we take advantage of next.



Condensing the exponential term

- The combined exponential argument becomes:

$$-\frac{1}{2} \left(\begin{bmatrix} x \\ z \end{bmatrix} - \begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix} \right)^T \Sigma_{\tilde{X}}^{-1} \left(\begin{bmatrix} x \\ z \end{bmatrix} - \begin{bmatrix} \bar{x} \\ \bar{z} \end{bmatrix} \right) + \frac{1}{2} (z - \bar{z})^T \Sigma_{\tilde{z}}^{-1} (z - \bar{z}).$$

- To condense notation somewhat, define $\tilde{x} = x - \bar{x}$ and $\tilde{z} = z - \bar{z}$.
- Then, the terms in the exponent become:

$$-\frac{1}{2} \begin{bmatrix} \tilde{x} \\ \tilde{z} \end{bmatrix}^T \Sigma_{\tilde{X}}^{-1} \begin{bmatrix} \tilde{x} \\ \tilde{z} \end{bmatrix} + \frac{1}{2} \tilde{z}^T \Sigma_{\tilde{z}}^{-1} \tilde{z}.$$



Inverting covariance (step 1)

- To proceed, we must invert $\Sigma_{\tilde{x}}$. To do so, we substitute the following transformation (which you can verify):

$$\begin{bmatrix} \Sigma_{\tilde{x}} & \Sigma_{\tilde{x}\tilde{z}} \\ \Sigma_{\tilde{z}\tilde{x}} & \Sigma_{\tilde{z}} \end{bmatrix} = \begin{bmatrix} I & \Sigma_{\tilde{x}\tilde{z}}\Sigma_{\tilde{z}}^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} \Sigma_{\tilde{x}} - \Sigma_{\tilde{x}\tilde{z}}\Sigma_{\tilde{z}}^{-1}\Sigma_{\tilde{z}\tilde{x}} & 0 \\ 0 & \Sigma_{\tilde{z}} \end{bmatrix} \begin{bmatrix} I & 0 \\ \Sigma_{\tilde{z}}^{-1}\Sigma_{\tilde{z}\tilde{x}} & I \end{bmatrix}.$$

- The terms highlighted in blue are the Schur complement of $\Sigma_{\tilde{x}}$:

$$M = \Sigma_{\tilde{x}} - \Sigma_{\tilde{x}\tilde{z}}\Sigma_{\tilde{z}}^{-1}\Sigma_{\tilde{z}\tilde{x}}.$$

- Then,

$$\begin{aligned} \begin{bmatrix} \Sigma_{\tilde{x}} & \Sigma_{\tilde{x}\tilde{z}} \\ \Sigma_{\tilde{z}\tilde{x}} & \Sigma_{\tilde{z}} \end{bmatrix}^{-1} &= \begin{bmatrix} I & 0 \\ \Sigma_{\tilde{z}}^{-1}\Sigma_{\tilde{z}\tilde{x}} & I \end{bmatrix}^{-1} \begin{bmatrix} M & 0 \\ 0 & \Sigma_{\tilde{z}} \end{bmatrix}^{-1} \begin{bmatrix} I & \Sigma_{\tilde{x}\tilde{z}}\Sigma_{\tilde{z}}^{-1} \\ 0 & I \end{bmatrix}^{-1} \\ &= \begin{bmatrix} I & 0 \\ -\Sigma_{\tilde{z}}^{-1}\Sigma_{\tilde{z}\tilde{x}} & I \end{bmatrix} \begin{bmatrix} M^{-1} & 0 \\ 0 & \Sigma_{\tilde{z}}^{-1} \end{bmatrix} \begin{bmatrix} I & -\Sigma_{\tilde{x}\tilde{z}}\Sigma_{\tilde{z}}^{-1} \\ 0 & I \end{bmatrix}. \end{aligned}$$



Inverting covariance (step 2)

- Substituting this result, the exponent is then:

$$\begin{aligned} \text{exponent} &= -\frac{1}{2}(\tilde{x}^T M^{-1} \tilde{x} - \tilde{x}^T M^{-1} \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} \tilde{z} - \tilde{z}^T \Sigma_{\tilde{z}}^{-1} \Sigma_{\tilde{z}\tilde{x}} M^{-1} \tilde{x} \\ &\quad + \tilde{z}^T \Sigma_{\tilde{z}}^{-1} \Sigma_{\tilde{z}\tilde{x}} M^{-1} \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} \tilde{z} + \tilde{z}^T \Sigma_{\tilde{z}}^{-1} \tilde{z} - \tilde{z}^T \Sigma_{\tilde{z}}^{-1} \tilde{z}). \end{aligned}$$

- The last two terms cancel; the remaining terms can be grouped as:

$$\text{exponent} = -\frac{1}{2}(\tilde{x} - \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} \tilde{z})^T M^{-1} (\tilde{x} - \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} \tilde{z}).$$

- In terms of only the original variables, the exponent is:

$$-\frac{1}{2} \left(x - (\bar{x} + \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} (z - \bar{z})) \right)^T \left(\Sigma_{\tilde{x}} - \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} \Sigma_{\tilde{z}\tilde{x}} \right)^{-1} \left(x - (\bar{x} + \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} (z - \bar{z})) \right).$$



A solution to the generic conditional Gaussian

- Comparing this exponent to Gaussian standard form:

$$-\frac{1}{2} \left(x - (\bar{x} + \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} (z - \bar{z})) \right)^T \left(\Sigma_{\tilde{x}} - \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} \Sigma_{\tilde{z}\tilde{x}} \right)^{-1} \left(x - (\bar{x} + \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} (z - \bar{z})) \right),$$

- Infer that mean of $f(x | z)$ must be $\bar{x} + \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} (z - \bar{z})$.
- Infer that covariance of $f(x | z)$ must be $\Sigma_{\tilde{x}} - \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} \Sigma_{\tilde{z}\tilde{x}}$.

- So, we conclude that:

$$\begin{aligned} f(x | z) &\sim \mathcal{N}(\bar{x} + \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} (z - \bar{z}), \Sigma_{\tilde{x}} - \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} \Sigma_{\tilde{z}\tilde{x}}) \\ \mathbb{E}[x | z] &= \bar{x} + \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} (z - \bar{z}) = \mathbb{E}[x] + \Sigma_{\tilde{x}\tilde{z}} \Sigma_{\tilde{z}}^{-1} (z - \mathbb{E}[z]). \end{aligned}$$



State update equation

- Applying this to our problem, when $z_k = \tilde{z}_k + \hat{z}_k$, we get:

$$\begin{aligned}\mathbb{E}[\tilde{x}_k^- | z_k] &= \mathbb{E}[\tilde{x}_k^-] + \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1} (z_k - \mathbb{E}[z_k]) \\ &= \mathbb{E}[\tilde{x}_k^-] + \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1} (\tilde{z}_k + \hat{z}_k - \mathbb{E}[\tilde{z}_k + \hat{z}_k]) \\ &= 0 + \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1} (\tilde{z}_k + \hat{z}_k - (0 + \hat{z}_k)) = \underbrace{\Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1}}_{L_k} \tilde{z}_k.\end{aligned}$$

- Putting all of the pieces together, we get the general update equation:

$$\hat{x}_k^+ = \hat{x}_k^- + L_k \tilde{z}_k.$$



Uncertainty of state estimate

- State-estimate covariance (uncertainty) $\Sigma_{\tilde{x},k}^+$ is:

$$\begin{aligned}\Sigma_{\tilde{x},k}^+ &= \mathbb{E}[(x_k - \hat{x}_k^+)(x_k - \hat{x}_k^+)^T] = \mathbb{E}[\{(x_k - \hat{x}_k^-) - L_k \tilde{z}_k\} \{(x_k - \hat{x}_k^-) - L_k \tilde{z}_k\}^T] \\ &= \mathbb{E}[\{\tilde{x}_k^- - L_k \tilde{z}_k\} \{\tilde{x}_k^- - L_k \tilde{z}_k\}^T] \\ &= \Sigma_{\tilde{x},k}^- - L_k \underbrace{\mathbb{E}[\tilde{z}_k (\tilde{x}_k^-)^T]}_{\Sigma_{\tilde{z},k} L_k^T} - \underbrace{\mathbb{E}[\tilde{x}_k^- \tilde{z}_k^T]}_{L_k \Sigma_{\tilde{z},k}} L_k^T + L_k \Sigma_{\tilde{z},k} L_k^T \\ &= \Sigma_{\tilde{x},k}^- - L_k \Sigma_{\tilde{z},k} L_k^T.\end{aligned}$$

- Final output of Gaussian sequential probabilistic inference has two parts:

- State estimate. Best guess of present state value, $\hat{x}_k^+$
- Covariance estimate. Uncertainty of $\hat{x}_k^+$, $\Sigma_{\tilde{x},k}^+$ can yield error bounds



Summary

Generic Gaussian sequential probabilistic inference recursion:

$$\begin{aligned}\hat{x}_k^+ &= \hat{x}_k^- + L_k (z_k - \hat{z}_k) = \hat{x}_k^- + L_k \tilde{z}_k \\ \Sigma_{\tilde{x},k}^+ &= \Sigma_{\tilde{x},k}^- - L_k \Sigma_{\tilde{z},k} L_k^T,\end{aligned}$$

where:

$$\begin{aligned}\hat{x}_k^- &= \mathbb{E}[x_k | \mathbb{Z}_{k-1}] & \Sigma_{\tilde{x},k}^- &= \mathbb{E}[(x_k - \hat{x}_k^-)(x_k - \hat{x}_k^-)^T] = \mathbb{E}[(\tilde{x}_k^-)(\tilde{x}_k^-)^T] \\ \hat{x}_k^+ &= \mathbb{E}[x_k | \mathbb{Z}_k] & \Sigma_{\tilde{x},k}^+ &= \mathbb{E}[(x_k - \hat{x}_k^+)(x_k - \hat{x}_k^+)^T] = \mathbb{E}[(\tilde{x}_k^+)(\tilde{x}_k^+)^T] \\ \hat{z}_k &= \mathbb{E}[z_k | \mathbb{Z}_{k-1}] & \Sigma_{\tilde{z},k} &= \mathbb{E}[(z_k - \hat{z}_k)(z_k - \hat{z}_k)^T] = \mathbb{E}[(\tilde{z}_k)(\tilde{z}_k)^T] \\ L_k &= \mathbb{E}[(x_k - \hat{x}_k^-)(z_k - \hat{z}_k)^T] \Sigma_{\tilde{z},k}^{-1} = \Sigma_{\tilde{x}\tilde{z},k}^- \Sigma_{\tilde{z},k}^{-1}.\end{aligned}$$

- Note that this is a linear recursion, even if the system is nonlinear(!)